

Comparative Analysis of Regularized and Tree-Based Machine Learning Models for Customer Churn Prediction

Based on the uploaded project: Customer Churn Prediction Using Elastic Net Regularized Logistic Regression

Abstract

Customer churn prediction can be approached using statistical and machine learning classifiers with different trade-offs between interpretability and predictive flexibility. This paper proposes a comparative experimental framework for Elastic Net logistic regression, plain logistic regression, decision trees and random forests. The study follows a common preprocessing pipeline and evaluates all models using accuracy, precision, recall, F1-score and ROC-AUC. Elastic Net is expected to be particularly useful for correlated customer variables because it combines L1 feature selection with L2 coefficient stabilization. Tree-based models provide complementary benchmarks for nonlinear relationships. The comparison is intended to support model selection according to both predictive performance and business interpretability.

Introduction

The source project notes that standard logistic regression can become unstable when customer features are numerous and collinear, whereas purely complex models may reduce interpretability. A comparative study therefore helps determine whether the additional regularization of Elastic Net provides a useful balance.

Experimental Design

Customer data are preprocessed consistently across all models. Missing values are handled, categorical variables encoded and numerical variables standardized where appropriate. A training/testing split is used. Elastic Net hyperparameters λ and α are selected by cross-validation. Plain logistic regression, decision tree and random forest classifiers are trained as benchmarks. All models are evaluated on the same held-out test data.

Evaluation Metrics

Accuracy measures overall classification correctness. Precision measures the proportion of predicted churners who actually churn. Recall measures the proportion of actual churners detected by the model. F1-score balances precision and recall, while ROC-AUC evaluates discrimination across classification thresholds. Reporting all five measures avoids relying on a single performance statistic.

Discussion

Elastic Net offers an explicit feature-selection mechanism and coefficient stability, making its predictions easier to communicate. Decision trees can provide intuitive decision rules, while random forests can capture more complex patterns. The final model should therefore be selected using both measured predictive performance and the organization's need for explanation and deployment.

Conclusion

A systematic comparison of regularized and tree-based models provides a stronger basis for customer churn model selection than accuracy alone. The proposed experiment directly reflects the source project's benchmarking strategy and can be implemented with the Telco Customer Churn dataset.

References / Source Basis

1. Uploaded project presentation, "Customer Churn Prediction Using Elastic Net Regularized Logistic Regression," A.Y. 2026–27. The presentation describes the business problem, Elastic Net rationale, methodology pipeline, evaluation metrics, Telco Customer Churn dataset and related literature.
2. Arivazhagan, B. & Sankara Subramanian, R.S. "Customer Churn Prediction using Logistic Regression with Regularization and Optimization Technique." IJITEE, Vol. 9, Issue 9, 2020.
3. Šimović, P. P., Chen, C. Y. T. & Sun, E. W. "Classifying the Variety of Customers' Online Engagement for Churn Prediction with a Mixed-Penalty Logistic Regression." Computational Economics, 2023.
4. Yang, L., Li, D. & Lu, Y. "Prediction Modeling and Analysis for Telecom Customer Churn in Two Months." arXiv:1911.00558, 2019.